Diophantine Equations

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1 Examples

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Example 1: $a; b; c; d \in \mathbb{Z}^+$. Solve the equation:

$$2^{a!} + 2^{b!} + 2^{c!} = d^3$$

Proof: A slightly different solution.

If $a, b, c \geq 3$, $d^3 \equiv 3 \pmod 7$, $\Rightarrow\Leftarrow$.

WLOG $a \geq b \geq c$. $2^2(2^{a!-2} + 2^{b!-2} + 1) = d^3$. Thus, $b! = 2 \implies b = 2$.

$2^{a!} + 8 = d^3$. There are no perfect cubes with difference 8, so $a \leq 2 \implies a = 2$.

However, $2^2 + 2^2 + 2^2 = 12 \neq d^3$ for any $d \in \mathbb{Z}_+$.

Thus, $c = 1$. $2(2^{a!-1} + 2^{b!-1} + 1) = d^3 \implies b! - 1 = 0$.

$4(2^{a!-2} + 1) = d^3 \implies a = 2$.

Thus, $(a, b, c, d) = (2, 1, 1, 2), (1, 1, 2, 2), (1, 2, 1, 2)$.

AoPS solution: me

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Example 2: Determine all positive integers a, b and c such that
$2 \cdot 10^a + 5^b = 2025^c$

Proof: Taking v_5 on both sides gives $\min\{a, b\} = 2c$.

Case 1: $b = 2c$

$2 \cdot 10^a = 2025^c - 25^c$

Since $2025^1 - 25^1 = 2 \cdot 10^3$, this is the only solution by Zsig.

Case 2: $a = 2c$

$2 \cdot 10^{2c} + 5^b = 2025^c$

$5^b = 5^{2c}(81^c - 2^{2c+1})$

Since $2^{2c+1} \not\equiv 1 \pmod 5$,

$b = 2c$ which is just Case 1.

So, the only solution is $(a, b, c) = (3, 2, 1)$.

AoPS solution: me

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Example 3: Find all pairs of primes (p, q) for which $p - q$ and $pq - q$ are both perfect squares.

Proof: Let $pq - q = a^2$, $p - q = b^2$.

Subtracting gives $p(q-1) = (a+b)(a-b)$. Since $a - b < a < p$ and $a + b < 2a < 2p$, we get $a + b = p$ and $a - b = q - 1$.

Finally, $p > q$, so $q = 2 \implies a = \frac{p+q-1}{2} = \frac{p+1}{2} \implies 8p - 8 = p^2 + 2p + 1$, so $p = 3$.

So, the only pair $(p, q) = (3, 2)$.

AoPS solution: me

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Example 4: Determine all prime numbers p and all positive integers x and y satisfying

$$x^3 + y^3 = p(xy + p).$$

Proof: Assume for the sake of contradiction $x + y = kp$.

If $k = 1$, $(x - y)^2 = p$, which is not possible.

If $k \geq 2$, $\frac{x+y}{k} + xy = p + xy \geq k(x^2 - xy + y^2) \geq 2xy$, so $\frac{x+y}{k} \geq xy$, $\Rightarrow\Leftarrow$.

Let $x + y = m$ and $xy = n$, so $\gcd(p, m) = 1$ and $m(m^2 - 3n) = p(n + p)$.

$\iff m^3 - p^2 = n(3m + p)$

$\iff m^3 - p^2 \equiv 0 \pmod{3m + p}$.

Now, notice $m^3 - p^2 \equiv m^3 - p^2 + 3mp + p^2 \equiv m(m^2 + 3p) \pmod{3m + p}$.

So, $3m + p \mid m^2 + 3p \iff 3m + p \mid 3m^2 + 9p - 3m^2 - pm \iff 3m + p \mid 9 - m$.

Clearly, $m = 9$. From here, checking just 4 cases gives the solutions $(p, x, y) = (8, 1, 19), (7, 2, 13), (5, 4, 7)$ and permutations.

AoPS solution: me

2 Practise Problems

Here are some practise problems, in increasing order of difficulty.

1. **IMO 2006 P4**

   Determine all pairs (x, y) of integers such that

   $$1 + 2^x + 2^{2x+1} = y^2.$$

2. **IMO 2015 P2**

   Determine all triples (a, b, c) of positive integers such that each of the numbers

   $$ab - c, \qquad bc - a, \qquad ca - b$$

   is a power of 2. (A power of 2 is an integer of the form 2^n, where n is a non-negative integer.)

3. **IMO 1988 P6**

   Let a and b be positive integers such that $ab + 1$ divides $a^2 + b^2$. Show that

   $$\frac{a^2 + b^2}{ab + 1}$$

   is the square of an integer.